Finding and Verifying Inverse Functions
Answer Key

Algebra 2 Composition, Inverses and Transformations

Quick Answers

1. $x - 7$
2. $\frac{x}{5}$
3. $\frac{x}{3} + \frac{8}{3}$, x
4. x, x
5. 3
6. $5x + 4$
7. $x^2 + 2$
8. x, x
9. 7
10. $\frac{3x+1}{x-2}$, x
11. -3, -3
12. 3

Curriculum

Level: Algebra 2 Composition, Inverses and Transformations
HSF-BF.A.1c, HSF-BF.B.4 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 18 tagged checks.

Common wrong answers

5. *If they got 43: evaluated f at 11 instead of the inverse, giving 43*
11. *If they got -6.33: divided before subtracting, giving about -6.33 instead of -3*

1. Inverse of $f(x) = x + 7$.

Step 1. Write $y = x + 7$, then swap the letters: $x = y + 7$. The swap is what turns "output from input" into "input from output".
Step 2. Solve for y: subtract 7 from both sides, giving $y = x - 7$.
Step 3. $f^{-1}(x) = \boxed{x - 7}$
Check. $f(f^{-1}(x)) = (x - 7) + 7 = x$ ✓

2. Inverse of $f(x) = 5x$.

Step 1. $y = 5x$ becomes $x = 5y$ after the swap.
Step 2. Divide both sides by 5: $y = \dfrac{x}{5}$.
Step 3. $f^{-1}(x) = \boxed{\dfrac{x}{5}}$
Check. $f(f^{-1}(x)) = 5 \cdot \dfrac{x}{5} = x$ ✓
Listen for: $f^{-1}(x) = \frac{1}{5x}$ — that is the reciprocal of f, not the inverse function.

3. Inverse of $f(x) = 3x - 8$.
Step 1. $y = 3x - 8$; swap to get $x = 3y - 8$.
Step 2. Undo in reverse order: add 8 to get $x + 8 = 3y$, then divide by 3.
Step 3. $f^{-1}(x) = \boxed{\dfrac{x+8}{3}}$
Check. $f\left(\dfrac{x+8}{3}\right) = 3 \cdot \dfrac{x+8}{3} - 8 = (x+8) - 8 = \boxed{x}$
Common wrong answer: $\frac{x}{3} + 8$ — the 8 was moved after the division instead of before it.

4. Verifying $f(x) = 2x + 9$ and $g(x) = \dfrac{x-9}{2}$.
Step 1. A pair is inverse only if *both* compositions return the input untouched, so compute each one.
Step 2. $f(g(x)) = 2 \cdot \dfrac{x-9}{2} + 9 = (x-9) + 9 = \boxed{x}$
Step 3. $g(f(x)) = \dfrac{(2x+9)-9}{2} = \dfrac{2x}{2} = \boxed{x}$
Both directions give x, so f and g are inverses.
Why both: one-sided composition can equal x on a restricted set while the pair still fails elsewhere. Checking both is the definition, not extra caution.

5. $f^{-1}(11)$ for $f(x) = 4x - 1$.
Step 1. Find the formula first: $x = 4y - 1$ gives $y = \dfrac{x+1}{4}$, so $f^{-1}(x) = \dfrac{x+1}{4}$.
Step 2. Evaluate at 11: $\dfrac{11+1}{4} = \dfrac{12}{4}$.
Step 3. $f^{-1}(11) = \boxed{3}$
Check. $f(3) = 4(3) - 1 = 11$ ✓
Common wrong answer: 43 — f was evaluated at 11 instead of f^{-1}.

6. Inverse of $f(x) = \dfrac{x-4}{5}$.
Step 1. Swap: $x = \dfrac{y-4}{5}$.
Step 2. Multiply both sides by 5: $5x = y - 4$, then add 4.
Step 3. $f^{-1}(x) = \boxed{5x + 4}$
Check. $f^{-1}(f(x)) = 5 \cdot \dfrac{x-4}{5} + 4 = (x-4) + 4 = x$ ✓

7. Inverse of $f(x) = \sqrt{x-2}$, domain $x \geq 2$.
Step 1. Swap: $x = \sqrt{y-2}$.
Step 2. Square both sides: $x^2 = y - 2$, so $y = x^2 + 2$.
Step 3. $f^{-1}(x) = \boxed{x^2 + 2}$
Step 4 (the domain). A square root never returns a negative number, so the outputs of f — and therefore the inputs of f^{-1} — are exactly $x \geq 0$. The rule $x^2 + 2$ makes sense for every real number, but as the inverse of this f its domain is $x \geq 0$. Domain and range swap places.
Listen for: an inverse quoted as "all reals". The formula and the function are not the same object; the inverse inherits its domain from the range of f.

8. Testing Sofia's claim: $f(x) = \dfrac{x+6}{2}$, $g(x) = 2x - 6$.

Step 1. Compose in one order: $f(g(x)) = \dfrac{(2x-6)+6}{2} = \dfrac{2x}{2} = \boxed{x}$

Step 2. Compose in the other order: $g(f(x)) = 2 \cdot \dfrac{x+6}{2} - 6 = (x+6) - 6 = \boxed{x}$

Step 3. Both simplify to x, so Sofia is right: the two functions are inverses.

Note: g undoes "add 6, then halve" by "double, then subtract 6" — the operations reverse and so does their order.

9. $f(x) = x^2$ on $x \geq 0$, and $f^{-1}(49)$.

Step 1. With the domain cut to $x \geq 0$, each output comes from exactly one input, so an inverse exists. Swapping gives $x = y^2$ with $y \geq 0$.

Step 2. Taking the non-negative root, $f^{-1}(x) = \sqrt{x}$, and $\sqrt{49} = 7$.

Step 3. $f^{-1}(49) = \boxed{7}$

Why the restriction matters. Without it both 7 and -7 square to 49, so "the" input is not defined and f has no inverse function at all.

10. Inverse of $f(x) = \dfrac{2x+1}{x-3}$.

Step 1. Swap: $x = \dfrac{2y+1}{y-3}$, then clear the fraction: $x(y-3) = 2y+1$.

Step 2. Expand and gather every y term on one side: $xy - 3x = 2y + 1$, so $xy - 2y = 3x + 1$.

Step 3. Factor out y — this is the step the whole problem turns on: $y(x-2) = 3x+1$, so $y = \dfrac{3x+1}{x-2}$.

Step 4. $f^{-1}(x) = \boxed{\dfrac{3x+1}{x-2}}$

Check. Substituting into f: $\dfrac{2\left(\frac{3x+1}{x-2}\right)+1}{\left(\frac{3x+1}{x-2}\right)-3} = \dfrac{\frac{6x+2+x-2}{x-2}}{\frac{3x+1-3x+6}{x-2}} = \dfrac{7x}{7} = \boxed{x}$ ✓

Note: the vertical asymptote of f is $x = 3$ and that of f^{-1} is $x = 2$ — the excluded input of one is the excluded output of the other.

11. $f^{-1}(-4)$ for $f(x) = 3x + 5$, two ways.

Step 1 (formula). $x = 3y+5$ gives $y = \dfrac{x-5}{3}$, so $f^{-1}(x) = \dfrac{x-5}{3}$ and $f^{-1}(-4) = \dfrac{-4-5}{3} = \dfrac{-9}{3} = \boxed{-3}$

Step 2 (direct). Solve $f(x) = -4$: $3x + 5 = -4$, so $3x = -9$ and $x = \boxed{-3}$

Step 3. The two agree, which is the whole point: asking "f^{-1} of -4" and asking "which input gives -4" are the same question.

Common wrong answer: about -6.33 — dividing by 3 before subtracting 5. Undo the operations in reverse order.

12. $f(x) = x^2 - 4$. *Part (a) is an open explanation, flagged for manual review; the model wording is below.*
Step 1 (part a). f is not one-to-one on the reals: $f(3) = 5$ and $f(-3) = 5$, so the output 5 has two inputs and there is no rule that sends 5 back to a single number. Graphically, a horizontal line meets the parabola twice, so the reflection across $y = x$ is not a function.
Step 2 (part b). Restricting to $x \geq 0$ keeps only the right half of the parabola, which passes the horizontal-line test. Swapping gives $x = y^2 - 4$, so $y^2 = x + 4$ and, taking the non-negative root, $f^{-1}(x) = \sqrt{x+4}$ on $x \geq -4$.
Step 3. $f^{-1}(5) = \sqrt{5+4} = \sqrt{9} = \boxed{3}$
Check. $f(3) = 9 - 4 = 5$ ✓
Full credit: part (a) must say that two different inputs share an output (or invoke the horizontal-line test), not merely that "the graph is a parabola". Part (b) needs the restriction stated, the formula $\sqrt{x+4}$, and the value 3.